

1. Write the expression, $x + x^2 + x^3 + x^4 + \cdots + x^{20}$ in sigma notation.

2. Write out the following sum $\sum_{n=0}^3 (n+1)^2$.

Evaluate the geometric series or state that it diverges.

3. $\sum_{k=0}^{\infty} \frac{1}{4^k}$

4. $\sum_{m=0}^{\infty} \frac{(-3)^{2n}}{5^n}$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

5. $\lim_{n \rightarrow \infty} \left\{ \frac{(-1)^n}{n} \right\}$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive.

6. $\sum_{k=0}^{\infty} \frac{k}{3k+2}$

Use the Ratio Test to determine whether the following series converges, diverges, or state that the test is inconclusive.

7. $\sum_{k=1}^{\infty} \frac{1}{k!}$ (recall that $\frac{k!}{(k+1)!} = \frac{1}{k+1}$).

8. Let $f(x) = 3e^{-x}$. (Each part is worth 4 points.)

a. Find the Taylor polynomial of order $n = 2$ for $f(x)$ centered at 0.

b. Use the Taylor polynomial from above to approximate $3e^{0.1}$.

c. Compute the absolute error in the approximation assuming the exact value is given by a calculator.

Give a set of parametric equations that describes the curve.

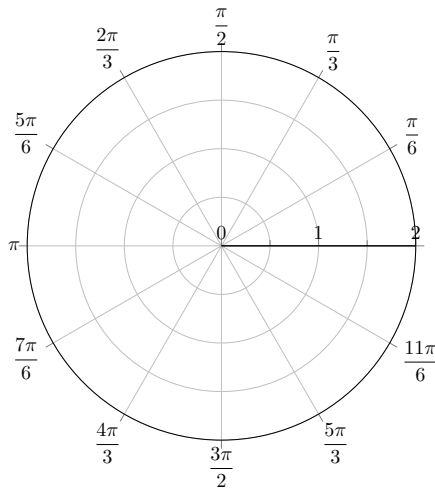
9. Give two different sets of parametric equations that generate a circle centered at $(0,0)$ with radius 5.

a. [2 pts] Set one:

b. [2 pts] Set two:

Locate the following polar coordinates on a graph.

10. A: $\left(1.5, \frac{\pi}{6}\right)$, B: $\left(1.5, \frac{\pi}{6} + \frac{\pi}{2}\right)$, C: $\left(1.5, -\frac{\pi}{6}\right)$, D: $\left(-1.5, -\frac{\pi}{6}\right)$



a.

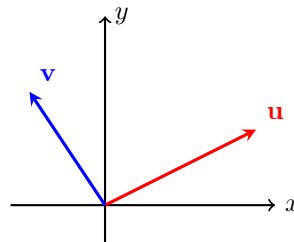
Figure 0.1: Use this graph to plot the polar coordinates.

Express the following **polar** coordinates in **Cartesian** coordinates.

11. $\left(2, \frac{\pi}{3}\right)$

Use the graph to draw the following vectors (provide labels). Make certain to indicate direction.

12. a. [2 pts] $\frac{1}{2}\mathbf{v}$
b. [2 pts] $\mathbf{u} + \mathbf{v}$



Use the vectors $\mathbf{u} = \langle -1, 1, 1 \rangle$ and $\mathbf{v} = \langle 2, -4, 0 \rangle$ to complete the following problems.

13. $\mathbf{u} + \mathbf{v}$

14. $\mathbf{u} \cdot \mathbf{v}$

15. Find the angle between $\mathbf{u}$ and $\mathbf{v}$. ($|\mathbf{u}| = \sqrt{3}$ and $|\mathbf{v}| = 2\sqrt{5}$.)

16. Find a vector perpendicular to $\mathbf{u}$ and $\mathbf{v}$.

17. A woman paddles a canoe at 5 mi/hr at 30 degrees ($\frac{1}{6}\pi$ radians) north of east. Find a vector describing the vertical and horizontal components of the canoe's velocity.

Find the indicated partial derivative of the following functions.

18. Find f_x and f_y for $f(x, y) = 5x \cdot \sin 2xy$.

19. Find the four second partial derivatives of $f(x, y) = 2x^5y^2 + 2x^2$.

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

20. dW/dt where $W = x^2 + y^2$, $x = \cos 2t$, and $y = \sin 2t$.

Compute the gradient of the following function and evaluate it at the given point P .

21. $f(x, y) = e^{2xy}$; $P(2, 3)$

Find a vector that points in a direction of no change in the function at P .

22. $\ln(1 + xy)$; $P(2, 3)$

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, or saddle point.

23. [8 pts] $g(x, y) = x^4 + y^4 - 16xy$ with critical points $(0, 0)$, $(2, 2)$, and $(-2, -2)$.

BONUS

Use Lagrange multipliers to find the maximum and minimum values of f (when they exist) subject to the constraint.

24. $f(x, y) = e^{2xy}$ subject to $x^3 + y^3 = 16$